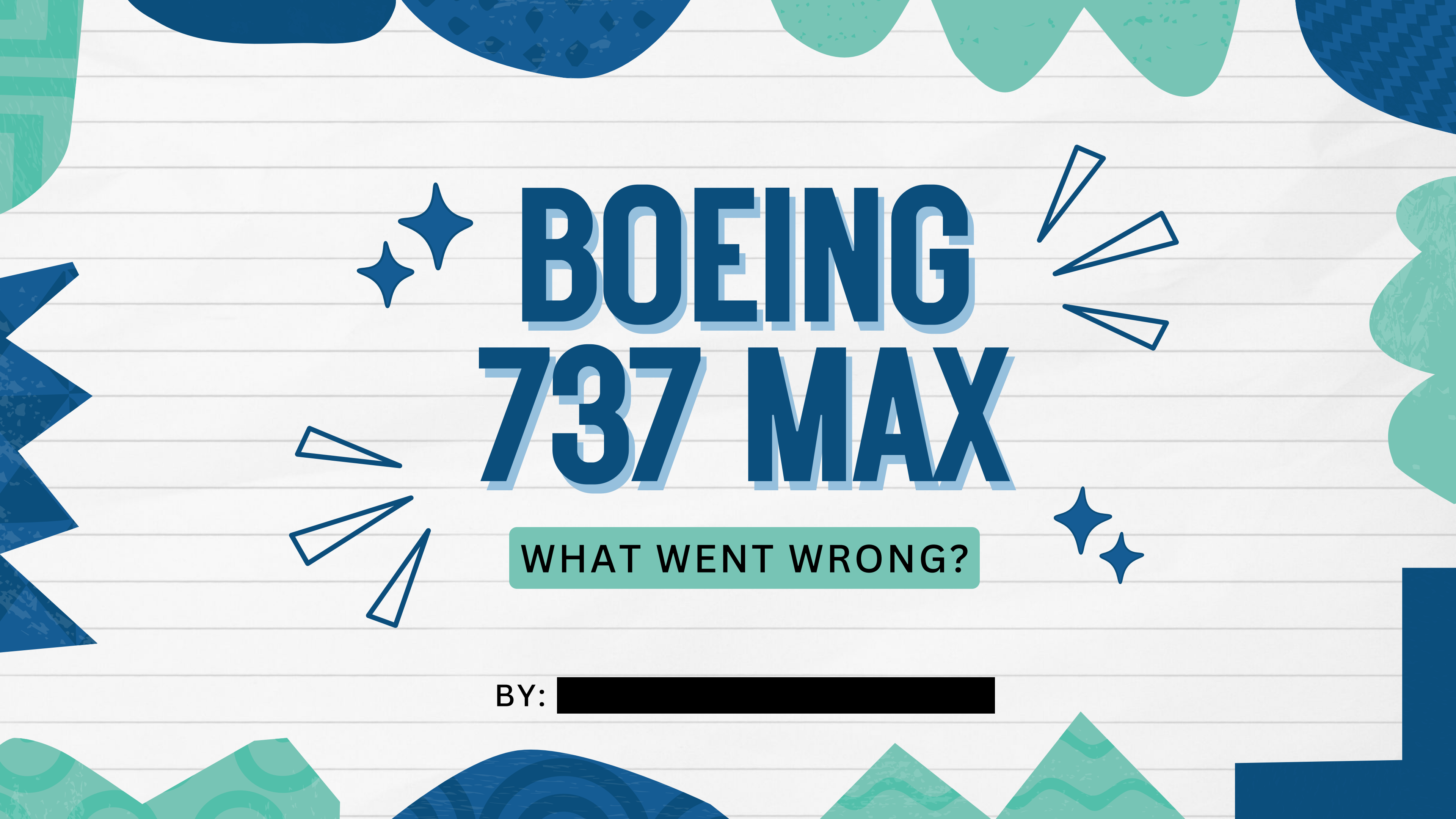




BOEING 737 MAX

WHAT WENT WRONG?

BY:



COMPUTER FAILURES

HARDWARE ERRORS

The MCAS only relied on one sensor, which had a fault.

SOFTWARE ERRORS

The MCAS was the main contribution to the two fatal incidents.

COMMUNICATION FAILURE

Pilots did not receive enough training to use the MCAS, resulting in human-computer miscommunication.

BIRSCH'S MORAL RESPONSIBILITY



1. **FIRST CRITERIA:** The MCAS that the programmers developed have caused 2 fatal incidents which caused the death of 346 lives. This confirms that the first criteria was met.
2. **SECOND CRITERIA:** While some of Boeing's employees have already knew of the issue with the MCAS, they have not taken any further actions to fix the issues. Furthermore, Boeing have also withheld crucial information from the FAA during the certification process. This confirms that the second criteria was met due to Boeing's negligence.
3. **THIRD CRITERIA:** Boeing knew about the faulty system, but they did not take any responsibility in fixing the system. Furthermore, Boeing did not issue any simulation training to pilots to save costs, while knowing the consequences. These information confirms that the thirs criteria was met.

INCIDENT PREVENTION

[BOEING]



- Issue a simulation training for pilots before debuting the 737 Max, instead of adopting an iPad training to cut costs.
- Take more time during the development and testing phase, instead of rushing to release the model due to competition.
- Implement an option to turn off the MCAS system.
- Consult with pilot testers and ensure that pilots are fully comfortable with the system before its release.

INCIDENT PREVENTION

[FAA]



- Take through steps to certify the new model.
- Require a simulation training for pilots for any new aircraft model.
- Ensure that all employees do not oversee any steps during the approval process on the behest of Boeing

WHO IS RESPONSIBLE?

- Those who are morally responsible are those in Boeing who knew about the aircraft faults.
- Those who are legally responsible are:
 - FAA for not carrying out the approval process properly
 - Technical and testing team of Boeing who knew about the faults and did not take any actions
 - Boeing officials who knew about the faults but choose to ignore them to meet with competition deadlines.

**THANK
YOU VERY
MUCH!**

